

***Proof of $2^n - 1$ steps through
Generating Function.***

Number of moves = h_n , required to solve the puzzle of `n` disks on the three rods is given by the recurrence relation.

$h_n = 2h_{n-1} + 1$ [i. e. $T(n) = 2T(n - 1) + 1$, which we will get recurrence tree after constructing the program.]

There fore, $h_{n+1} = 2h_{n-1+1} + 1 = 2h_n + 1$.

And $h_1 = 1$.

$$g(x) = \sum_{n=0}^{\infty} h_{n+1} x^n$$

Therefore recreating the generating function as:

$$= h_1 + \sum_{n=1}^{\infty} h_{n+1} x^n$$

$$= h_1 + \sum_{n=1}^{\infty} (2h_n + 1)x^n \quad [As, h_{n+1} = 2h_n + 1]$$

Adding $n + 1$ we get:

$$= h_1 + \sum_{n=1}^{\infty} (2h_{n+1} + 1)x^{n+1}$$

$$= h_1 + \sum_{n=1}^{\infty} (2h_{n+1} + 1)x^n \times x \quad [As, x^{n+1} = x \times x^n]$$

$$= h_1 + \sum_{n=1}^{\infty} (2h_{n+1}x^n + x^n)x \quad [As, x^{n+1} = x \times x^n]$$

$$= h_1 + \left(2 \sum_{n=1}^{\infty} h_{n+1}x^n + \sum_{n=1}^{\infty} x^n \right) x$$

$$= h_1 + \left(2g(x) + \frac{1}{1-x} \right) x \left[g(x) \right. \\ \left. = \sum_{n=1}^{\infty} h_{n+1}x^n \quad and \quad \sum_{n=1}^{\infty} x^n = \frac{1}{1-x} \right]$$

$$= h_1 + 2xg(x) + \frac{x}{1-x}$$

Now we know , $h_1 = 1$, hence we get:

$$= 1 + 2xg(x) + \frac{x}{1-x}$$

$$\therefore g(x) = 1 + 2xg(x) + \frac{x}{1-x}$$

$$\Rightarrow g(x) - 2xg(x) = 1 + \frac{x}{1-x}$$

$$\Rightarrow g(x)(1 - 2x) = \frac{1 - x + x}{1 - x}$$

$$\Rightarrow g(x)(1 - 2x) = \frac{1}{1 - x}$$

$$\Rightarrow g(x) = \frac{1}{(1 - x)(1 - 2x)}$$

The partial fraction can be written as follows: –

$$\Rightarrow g(x) = \frac{A}{1-2x} + \frac{B}{1-x}$$

$$\text{or, } \frac{1}{(1-2x)(1-x)} = \frac{A}{1-2x} + \frac{B}{1-x}$$

$$\text{or, } \frac{1}{(1-2x)(1-x)} = \frac{A(1-x) + B(1-2x)}{(1-2x)(1-x)}$$

$$\text{or, } 1 = A(1-x) + B(1-2x)$$

<i>Now,</i> $1-x=0$ $\Rightarrow -x=-1$	$1-2x=0$ $\Rightarrow -2x=-1$ $\Rightarrow x=\frac{1}{2}$
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<i>Hence,</i> $1 = A\left(1 - \frac{1}{2}\right)$ $\Rightarrow 1 = A\left(\frac{2-1}{2}\right)$ $\Rightarrow 1 = A\left(\frac{1}{2}\right)$	$1 = B(1 - 2 \times 1)$ $\Rightarrow 1 = B(1 - 2)$ $\Rightarrow 1 = -B$ $\Rightarrow B = -1.$
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$$\Rightarrow A = 2$$

Now, coming back to the partial fraction we get:

$$\begin{aligned}\therefore g(x) &= \frac{A}{1-2x} + \frac{B}{1-x} \\ &= \frac{2}{1-2x} - \frac{1}{1-x}\end{aligned}$$

As we know ,

$$\sum_{n=1}^{\infty} x^n = \frac{1}{1-x} \text{ and } \sum_{n=1}^{\infty} (2x)^n = \frac{1}{1-2x}$$

$$\therefore 2 \times \frac{1}{1-2x} - \frac{1}{1-x}$$

$$= 2 \times \sum_{n=1}^{\infty} (2x)^n - \sum_{n=1}^{\infty} x^n$$

$$= 2 \times \sum_{n=1}^{\infty} 2^n x^n - \sum_{n=1}^{\infty} x^n$$

$$= \sum_{n=1}^{\infty} 2^{n+1} x^n - \sum_{n=1}^{\infty} x^n [2 \times 2^n = 2^{n+1}]$$

$$= \sum_{n=1}^{\infty} (2^{n+1} - 1) x^n$$

such that $h_n = 2^n - 1$.

*Hence ,the minimal number of moves for the $n - \text{disk}$
Tower of Hanoi is $2^n - 1$.*
